

# CS 374A Review MT2

## Electric Boogaloo

ACM @ UIUC

April 11, 2026



# Disclaimers and Logistics

- This review session is being recorded. Recordings and slides will be distributed on EdStem after the end.
- **Agenda:** We'll quickly review all topics likely to be covered, then go through a practice exam, then review individual topics by request.
- Please let us know if we're going too fast/slow, not speaking loud enough/speaking too loud, etc.
- If you have a question anytime during the review session, please ask! Someone else almost surely has a similar question.
- We'll provide a feedback form at the end of the session.

# Recursion

- **Definition:** Reducing the problem to a smaller instance of itself, where eventually we can terminate in a base case.
  - Think: If we have a problem of size  $n$ , we want to continuously reduce to a problem smaller than  $n$ .
  - Example: Binary search, Karatsuba

## Template

```
1: procedure AMAZINGRECURSIVEALGO( $n$ )
2:   if  $n ==$  [some base case] then
3:     return [value]
4:   else
5:     return AmazingRecursiveAlgo( $n/2$ )
```

- Similar to **induction**!

# Recursion: Runtime Analysis

- **General Form:**

$$T(n) = \underbrace{r}_{\substack{\text{\# of subproblems}}} \cdot \overbrace{T\left(\frac{n}{c}\right)}^{\substack{\text{work at each subproblem}}} + \underbrace{f(n)}_{\substack{\text{work at current level}}}$$

- Describes how the amount of work changes between each level of recursion.
- We can solve for a **time complexity** that describes the scaling behavior of the algorithm at hand.

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- **Master Theorem**

## Master Theorem

Decreasing:  $r \cdot f(n/c) = \kappa \cdot f(n)$  where  $\kappa < 1 \implies T(n) = O(f(n))$

Equal:  $r \cdot f(n/c) = \kappa \cdot f(n)$  where  $\kappa = 1 \implies T(n) = O(f(n) \cdot \log_c n)$

Increasing:  $r \cdot f(n/c) = \kappa \cdot f(n)$  where  $\kappa > 1 \implies T(n) = O(n^{\log_c r})$

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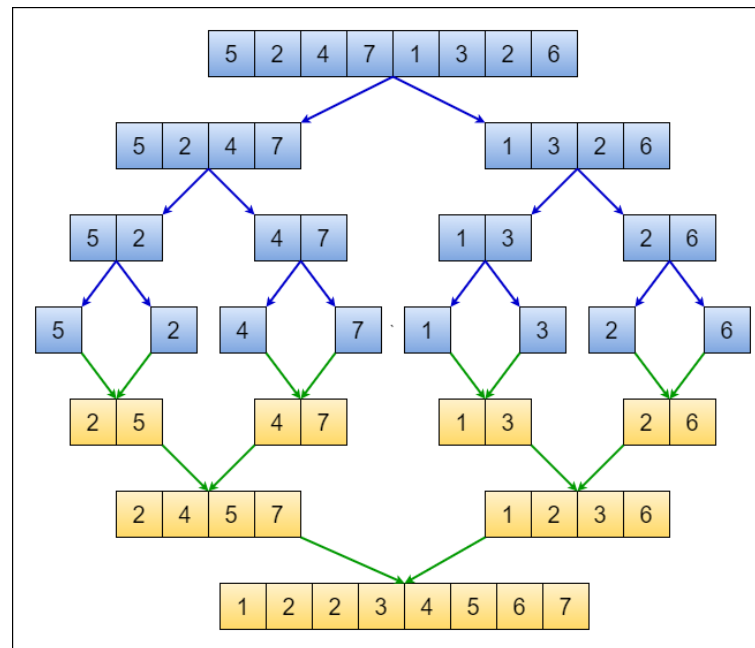
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- **Intuition:** If each level contains more work than the level below it, then the root level will dominate. If each level contains the same amount of work, then we have  $\log_c n$  levels with  $f(n)$  work. If each level contains less work than the work below it, then the leaf nodes will dominate.

# Divide and Conquer Algos: Merge Sort

- **Purpose:** Sort an arbitrary array.
- **Time Complexity:**  $O(n \log n)$
- **Intuition:** Three phases: (a) split the array in half, (b) sort each side, (c) merge the sorted halves by repeatedly comparing smallest elements on each side not yet inserted.

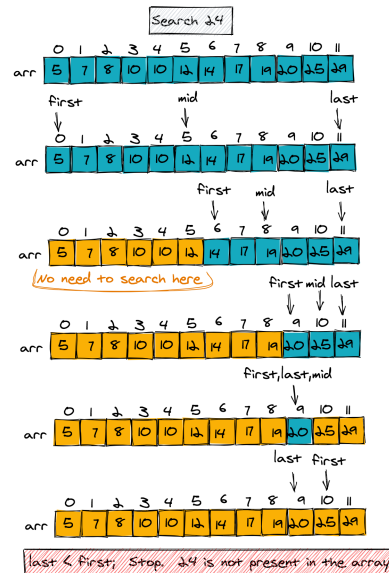


# Divide and Conquer Algos: MomSelect

- **Purpose:** Get the  $r^{\text{th}}$  smallest element in an arbitrary array.
- **Time Complexity:**  $O(n)$
- **Intuition:** Pick a pivot  $P$  with a value  $P_V$  and rearrange the array such that all the elements that are less than  $P_V$  are to the left of  $P$  and all the elements that are greater than  $P_V$  are to the right of  $P$ . If the length of the array of elements that are less than  $P_V$  is greater than  $r$ , then we know that the  $r^{\text{th}}$  smallest element is to the left of  $P$  and we recurse on the left subarray. Otherwise, we know that the  $r^{\text{th}}$  smallest element is to the right of  $P$  and we recurse on the right subarray.

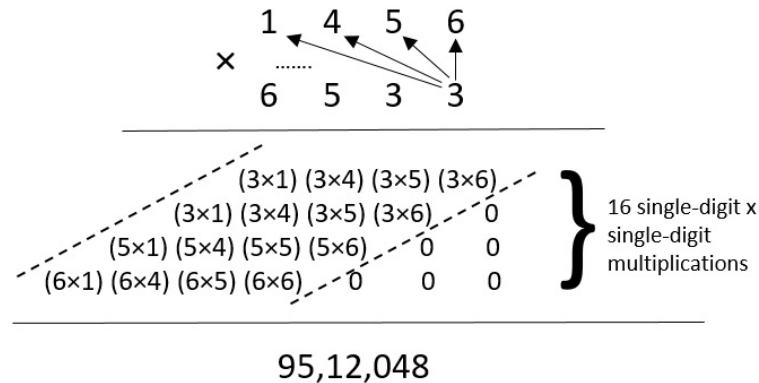
# Divide and Conquer Algos: Binary Search

- **Purpose:** Find the existence of an element in a sorted array
- **Time Complexity:**  $O(\log n)$
- **Intuition:** Say we are trying to find the value  $k$ . Pick the middle element  $M$  in the array. If  $k > M$ , the element must be to the right of  $M$  and we recurse on the right. Otherwise, we recurse on the left.



# Divide and Conquer Algos: Karatsuba's

- **Purpose:** Multiplication
- **Time Complexity:**  $O(n^{\log_2 3}) \approx O(n^{1.585})$
- 3 Phases:
  - **Divide:** Represent  $x$  as  $x_0p^n + x_1$ ,  $y$  as  $y_0p^n + y_1$
  - **Recurse:** Calculate  $x_0y_0$ ,  $x_1y_1$ ,  $(x_0 + x_1)(y_0 + y_1)$
  - **Combine:** Use the three results to calculate our final answer.



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- For a DP solution, we need:
  1. English Description
  2. Recurrence
  3. Memoization Structure
  4. Solution Location
  5. Evaluation Order
  6. Runtime

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  6. Runtime
- **How to solve a DP:**
  - Identify how we can take advantage of a recursive call on a smaller subset of the input space.
  - Identity base cases
  - Identity recurrences (they should cover all possible cases at each step)

# Dynamic Programming: Longest Increasing Subsequence

1. **English Description:**  $LIS(i, j)$  = the length of the longest increasing subsequence in  $A[i..n]$ , such that every element in the subsequence is  $> A[j]$ . Here,  $j = 0$  can be used as a sentinel with  $A[0] = -\infty$ .

2. **Recurrence:**

$$LIS(i, j) = \begin{cases} 0 & \text{if } i > n, \\ LIS(i + 1, j) & \text{if } A[i] \leq A[j], \\ \max(LIS(i + 1, j), 1 + LIS(i + 1, i)) & \text{if } A[i] > A[j]. \end{cases}$$

3. **Memoization Structure:** A 2D array  $LIS[0..n, 0..n]$ , storing all subproblem solutions  $LIS(i, j)$ .
4. **Solution Location:** The length of the LIS of the entire array is found at  $LIS(1, 0)$ , starting from the first element with the sentinel.
5. **Evaluation Order:** Fill the table in decreasing order of  $i$ .
6. **Runtime:** The table has  $O(n^2)$  entries, each computed in  $O(1)$ , so the total runtime is  $O(n^2)$ .

# Graphs

- **Definition:** A set of vertices  $V$  connected by a set of edges  $E$ . Individual edges are notated as  $(u, v)$ , where  $u, v \in V$ .
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- **Path:** A sequence of distinct vertices where each pair of consecutive vertices have an edge
- **Cycle:** A sequence of distinct vertices where each pair of consecutive vertices have an edge **and** the first and last vertices are connected.

# Graph Algorithms: Traversal

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- **BFS:**

- **Purpose:** Reachability, Shortest Path (unweighted graph)
- **Implementation details:** Add your neighbours to a **queue**, pop from the queue to get next node
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- **DFS:**

- **Purpose:** Reachability, toposort
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# Graph Algorithms: Shortest Path

- **Dijkstra's**

- **Purpose:** SSSP, no negative edges
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- **Bellman-Ford:**

- **Purpose:** SSSP, yes negative weights. Will detect negative cycles.
- **Implementation:** Dynamic Programming recurrence:
  - ▶  $d(v, k)$  is the shortest-walk distance from  $s$  to  $v$  using at most  $k$  edges
  - ▶  $d(v, k) = \min \left( d(v, k - 1), \min_{u \rightarrow v} d(u, k - 1) + \ell(u \rightarrow v) \right)$
- **Runtime:**  $O(mn)$

# Graph Algorithms: Longest Path

## Longest path in a Directed Acyclic Graph (DAG)

- **Purpose:** To find the longest simple path (no repeating vertices) by weight in a graph which is guaranteed to be a DAG<sup>1</sup>.
- **Runtime**<sup>2</sup>:  $O(V + E)$
- **Returns:** The sum of the weights of the longest path in the DAG.

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<sup>1</sup>Finding the longest path in other types of graphs is at least NP-hard.

<sup>2</sup>This is a relatively straight-forward DP on a DAG problem if you wish to derive it.

# Graph Problems: General Stuff

## How to solve graph problems:

1. Identify type of problem (Reachability, Shortest Path, SCC)
2. Construct new graph
  - Add sources/sinks
  - Add vertices via  $V' = V \times \{\text{some set}\}$  (Useful for tracking states)
  - Add vertices via  $E' = E \times \{\text{some set}\}$  (Useful for allowing/prohibit certain behaviour)
3. Apply some stock algorithm **(DO NOT MODIFY THE ALGORITHMS - MODIFY THE INPUTS!)**
4. Draw connection between how to result of the algorithm upon the new graph relates to the solution of the original question.

# Recurrences and Asymptotics

Give a tight asymptotic bound for each recurrence:

- $T(n) = 4 T(\frac{n}{2}) + n \log_2 n$
- $T(n) = T(\frac{3n}{4}) + T(\frac{n}{4}) + 5n$
- $T(n) = 9 T(\frac{n}{3}) + n^2$

Group the following functions s.t.  $f$  and  $g$  are in the same group if  $f(x) \sim \Theta(g(x))$ , and sort the groups by runtime:

- $n^n$
- $\log \log n$
- $374^n$
- $n!$
- $\log(n + n^{374})$
- $\log n^n$
- $n^{1.000001}$
- $2^n$
- $n \log n^5$
- $\frac{1}{\log_n 2}$

# Recurrences and Asymptotics

# Divide and Conquer

Suppose you have two arrays:  $A$  of length  $n_1$ , sorted in ascending order, and  $B$  of length  $n_2$ , sorted in descending order. You decide to concatenate them to form a new array  $C = A \cdot B$  for some application.

All the numbers are distinct. After concatenation, you no longer care about  $A$  and  $B$  individually, and you don't know  $n_1$  or  $n_2$ . You only have  $C$  and its size  $n = n_1 + n_2$ .

For example, if  $A = [1, 2, 6, 7]$  and  $B = [8, 4, 3]$ , then  $C = [1, 2, 6, 7, 8, 4, 3]$ .

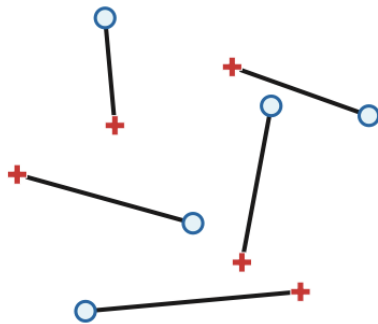
Given a number  $x$ , we want to check whether  $x$  appears in  $C$ .

# Divide and Conquer

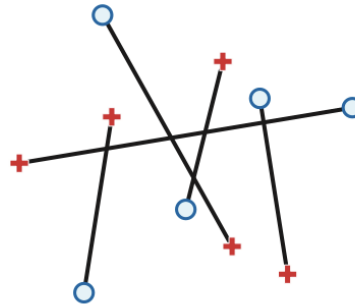
# Divide and Conquer

Given sets  $R, B \subset \mathbb{R}^2$  ( $n$  red points and  $n$  blue points) in the plane, we define

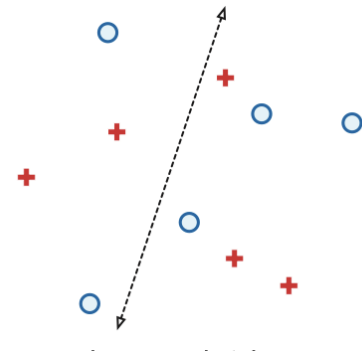
- **Tidy Matching:**  $n$  non-intersecting segments pairing each red point to a unique blue point. (Fig. 1(a) vs 1(b))
- **Ham-sandwich cut  $\ell$ :** A line with exactly  $\lfloor n/2 \rfloor$  red and  $\lfloor n/2 \rfloor$  blue points strictly above it. (Fig. 1(c))



(a) Tidy Matching



(b) Not a tidy matching



(c) Ham-Sandwich Cut

*Fact: A valid tidy matching and ham-sandwich cut always exist.*

## Black-Box Subroutines:

- $\text{HamSammy}(R, B)$  : Returns a ham-sandwich cut  $\ell$  in  $O(n)$  time.
- $\text{Above?}(p, \ell)$  : Returns `True` if  $p$  is strictly above  $\ell$  in  $O(1)$  time.

Describe and analyze an algorithm to compute a Tidy Matching of  $R$  and  $B$ , using these subroutines.

# Divide and Conquer

# Dynamic Programming

A *shuffle* of two strings  $X$  and  $Y$  is formed by interspersing the characters into a new string, keeping the characters of  $X$  and  $Y$  in the same order. For example, the string **BANANAANANAS** is a shuffle of the strings **BANANA** and **ANANAS** in several different ways.

**BANANAANANAS**   **BANANAANANAS**   **BANANAANANAS**

Similarly, the strings **PRODGYRNAMAMMIINCG** and **DYPRONGARMAMMICING** are both shuffles of the strings **DYNAMIC** and **PROGRAMMING**:

**PRODGYRNAMAMMIINCG**   **DYPRONGARMAMMICING**

Given three strings  $A[1..m]$ ,  $B[1..n]$ , and  $C[1..m+n]$ , describe and analyze an algorithm to determine whether  $C$  is a shuffle of  $A$  and  $B$ .

# DP

# What *is* it about cephalopods?

An entrepreneurial octopus runs a "Fish Wash" at the local coral reef. At exactly 8:00 AM, a morning rush of  $n$  fish arrive to be cleaned.

To handle the rush, the octopus acts as a multi-lane car wash, dedicating exactly **3 of its arms** to scrub the fish.

Each fish  $i$  (for  $1 \leq i \leq n$ ) requires  $\ell_i$  minutes to scrub clean, where  $\ell_i$  is a positive integer at most  $L$ . The octopus assigns each fish to one of the 3 arms. Fish assigned to the same arm form a line and must be cleaned in their original sequential order.

Describe an algorithm to assign the fish to the 3 arms in a way that **minimizes the sum of the waiting times** for all the fish. Analyze the runtime of your algorithm in terms of  $n$  and  $L$ .

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**Example:**  $\ell = [5, 3, 10, 20, 6, 15]$  and  $L = 20$ . Suppose the octopus assigns fish 1 and 3 to Arm 1; fish 2, 4, and 5 to Arm 2; and fish 6 to Arm 3.

- Fish 3 waits 5 mins (while Arm 1 cleans Fish 1).
- Fish 4 waits 3 mins (while Arm 2 cleans Fish 2).
- Fish 5 waits  $3 + 20 = 23$  mins (while Arm 2 cleans Fish 2 and 4).

The total waiting time is  $5 + 3 + 23 = 31$  minutes.

# DP

# Dynamic Programming

The **StupidScript** language includes a binary operator **@** that computes the *average* of its two arguments. For example, the StupidScript code `print(3 @ 6)` would print **4.5**, because  $(3 + 6)/2 = 4.5$ .

Expressions like `3@7@4` that use the **@** operator more than once yield different results when they are evaluated in different orders:

$$(3@7)@4 = 5@4 = 4.5 \quad \text{but} \quad 3@(7@4) = 3@5.5 = 4.25$$

Here is a larger example:

$$\begin{aligned} (((((8@6)@7)@5)@3)@(0@9)) &= 4.5 \\ ((8@6)@(7@5))@((3@0)@9) &= 5.875 \\ (8@(6@(7@(5@(3@0))))@9 &= 7.890625 \end{aligned}$$

Your goal for this problem is to describe and analyze an algorithm to compute, given a sequence of integers separated by **@** signs, the **largest possible** value the expression can take by adding parentheses. Your input is an array  $A[1 \dots n]$  listing the sequence of integers.

For example, if your input sequence is  $[3, 7, 4]$ , your algorithm should return 4.5, and if your input sequence is  $[8, 6, 7, 5, 3, 0, 9]$ , your algorithm should return 7.890625. Assume all arithmetic operations (including **@**) can be performed exactly in  $O(1)$  time.

# DP

# Bad Apple

Jeff is falling into a Bad Apple rabbit hole on Youtube. He wants to get from the original **Bad Apple** video to the video **bad apple but it's a sorting algorithm**<sup>3</sup> by *only* watching videos that are recommended to him in sequence.

Formally, you are given a list of  $n$  videos  $V$  such that each  $v \in V$  has a watch time  $\tau(v)$  and list  $R(v) \subseteq V$  of sidebar recommended videos that can be watched after  $v$  is watched. Suppose video  $s \in V$  is the original **Bad Apple** video and  $t \in V$  is the **bad apple but it's a sorting algorithm** video. We call a sequence of videos starting from  $s$  and ending at  $t$  a *Terpsichorean marathon*.

- (a) Describe and analyze an algorithm to find a *Terpsichorean marathon* of minimum total watch time.
- (b) We call a video a *banger* if it is in some *Terpsichorean marathon* with minimum watch time. Design and analyze an efficient algorithm to find all the *banger* videos.

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<sup>3</sup>Yes, this is a real thing.

# Graphs

# Who's the number one spice?

Young Cardamom is bringing the flavor to the fish!<sup>4</sup> His dish needs to have both **HEAT** and **SWEET** spices to be complete.

The kitchen can be modeled as a DAG where:

- Vertices represent prep stations
- Each station is labeled either HEAT, SWEET, or NEUTRAL
- A directed edge  $(u, v)$  with weight  $f(u, v)$  means station  $u$  can send ingredients to station  $v$ , contributing flavor value  $f(u, v)$
- Young Cardamom starts at station  $s$  and must reach plating station  $t$

To achieve the #1 spice, Young Cardamom's path must visit **at least one HEAT station** and **at least one SWEET station**.

Design and analyze an algorithm to find a path from  $s$  to  $t$  that visits both required station types and maximizes the total flavor value.

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<sup>4</sup>Of course, HAB is bringing the flavor to the rice.

# Graphs

# Graphs

We are given a directed graph  $G = (V, E)$  with  $n$  vertices and  $m$  edges, where  $m \geq n$ . Each edge  $e \in E$  has a weight  $w(e) \geq 0$ , and all weights are distinct. Let  $s$  and  $t$  be two distinct vertices in  $V$ .

For a path  $\pi$  from  $s$  to  $t$ , we define its **bottleneck value** as

$$b(\pi) = \max_{e \in \pi} w(e),$$

i.e., the weight of the heaviest edge along  $\pi$ .

We want to find a path from  $s$  to  $t$  that **minimizes the bottleneck value**.

# Graphs

# Feedback

- Further questions in ACM (Siebel 1104) after this
- Please fill out the feedback form:  
`go.acm.illinois.edu/cs374a_mt2_feedback`

